MOHAMED FAYAZ S.

Civil Engineering Student | Aspiring Civil Engineer

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PROFESSIONAL SUMMARY

First-year B.E. Civil Engineering student at Velammal College of Engineering and Technology, Madurai, with a strong interest in modern construction technologies, sustainable infrastructure, and engineering problem-solving. Developing foundational skills in AutoCAD, CAD, Python, GitHub, and web development. Currently exploring innovative civil engineering concepts including self-healing concrete and GIS-based flood risk mapping. Motivated to continuously learn, build practical projects, and develop technical skills throughout the engineering journey.

EDUCATION

B.E. Civil Engineering, Velammal College of Engineering and Technology, Madurai | ***2026 - 2030 (Expected Graduation: 2030)***

- Relevant Coursework: CAD, AutoCAD

Higher Secondary Education | ***Computer-Maths Group, 88.0%***

Secondary Education | ***77.8%***

TECHNICAL SKILLS

- Programming: Python
- Engineering & Design: CAD, AutoCAD
- Tools: GitHub
- Web: Web Development Basics
- Core Skills: Problem Solving, Team Collaboration, Technical Learning

PROJECTS

Self-Heal Buildings | ***Concept Project - Self-Healing Concrete***

Explored the concept of self-healing concrete, a technology designed to automatically seal or repair suitable cracks after they form, with the goal of improving the durability and service life of concrete structures.

Problem Addressed:

- Concrete structures are vulnerable to cracking and water or chemical ingress.
- Cracking can contribute to reinforcement corrosion and structural deterioration.
- Conventional repair methods require regular maintenance and increase long-term costs.
- Self-healing concrete aims to reduce maintenance requirements by enabling autonomous crack sealing.

GIS-Based Flood Risk Mapping | ***Concept Project - Geographic Information Systems***

Explored the development of a GIS-based digital mapping system for identifying and assessing areas that are more vulnerable to urban flooding by combining geographical and infrastructure-related factors.

Problem Addressed:

- Urban flooding can affect infrastructure, transportation, drainage systems, and communities.
- Identifying high-risk areas can support better drainage and infrastructure planning.
- GIS can combine multiple geographical factors to visualize and assess flood risk.
- The concept aims to support improved flood preparedness and infrastructure management.

PROFESSIONAL INTERESTS

- Sustainable Construction
- Structural Engineering
- Self-Healing Concrete
- GIS & Smart Infrastructure
- Flood Risk Management
- Construction Technology
- Engineering Design & CAD
- Web & Technology for Civil Engineering

LANGUAGES

English | Tamil

ADDITIONAL

- Building an engineering portfolio and GitHub presence
- Developing technical skills through self-learning and projects
- Interested in combining Civil Engineering + Technology to solve real-world infrastructure problems